2024 IEEE 2nd International Conference on Innovations in High Speed Communication and Signal Processing (IHCSP) | 979-8-3503-6894-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/IHCSP63227.2024.10960252

A Study on Deepfake Detection Methods

Shivnarayan Ahirwar[1*], Alpana Pandey[1]

[1]*Department of Electronics and Communication Engineering, Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh, India*

* Corresponding author E-mail shivnarayan.ahirwar@gmail.com (Shivnarayan Ahirwar)

Abstract

The continued development of deepfake Generative Adversarial Networks (GANs) have generated much controversy within the multiple fields including politics, entertainment, and healthcare. Deepfakes cause the generation of highly convincing fake news to spread with the aim of influencing public opinions, and given that the pictures are near realistic and the sound quality is excellent, it gets hard to differentiate between original and fake content. The release of apps for creating deepfakes has greatly increased the need to find ways to mitigate them because the consequences are criminal on the levels of fake news and privacy infringement. For instance, deepfake technology is capable of manipulating political messages, deceiving the electorate, and inaccurately portraying diseases with likelyhoods of wrong diagnosis and subsequent wrong treatment. This present paper presents a comprehensive survey and analysis of the existing deepfake detection methods including the traditional machine learning techniques, state of the art deep learning methods, and the recent approach based on blockchain technology. Unlike some previous literature reviews that may only include new literature from the last few years, this research synthesizes results from more recent findings to explicate system limitations and recent developments. The presented research is designed to provide methodological approaches and findings to the discussion on deepfake detection to facilitate further empirical work and practical developments beneficial for the crucial field. For deepfake technology lies on the path of digital media's continuous growth, therefore critical detection methods must be enhanced and united to maintain the desired information's purity and shield society against potential risks.

Keywords- Deepfake, CNN, GAN, Image Forensics.

Introduction

Over the last few years, deepfake technology has become a cause for major concern in many industries and concentrations such as political, entertainment, and healthcare industries. Deepfakes using second-generation GAN to generate realistic fake media can deceive the audiences and control the perception [1]. Despite the mirco-forensic chances that the utilized deep learning tools have enabled a burgeoning of deepfakes, it has become difficult to differentiate between real and fake contents [2-3]. Informing, misinforming and the privacy violation aspect of deep fake technology is worrisome to have around. For instance, deep fake videos have been deployed to spread fictions that could help shape a society's perception of certain political personalities. Further, in the medical domain, unscrupulous use of deepfake images of patients aggravates possible dangers such as wrong diagnose and mistreatment of the patients [4]. Therefore, there is the need for strong detection systems to be put in place to avoid such risks while protecting information [5].

The purpose of this paper is to present an overview of the state of the art of deepfake detection techniques, evaluating the issues encountered by existing frameworks and the progress that has been made in recent years. The following outlines various detection techniques: traditional ML, DL and new methods such as blockchain[6].

Thus, based on synthesising findings derived from the studied body of literature accruing in the recent past, this research aims to enrich the existing discussion on deepfake detection and may help propose a suite of directions for future research and potential practical consideration in this important area of study [7]. As the terrain of digital media changes, the need to detect these miscreants becomes more critical and this is where more research efforts, technologies and policies needs to be urgently developed.

Literature Review

The detection of deepfake videos has been occasioned by the fast-growing technologies and an increased worry of fake media. This literature review synthesizes findings from various studies, categorizing them into four primary approaches: Deep learning learning based methods, traditional AI approaches, statistical algorithms and solutions based on blockchain.

1. Artificial Neural Network Based Techniques

Deep learning has proved to be the best in the current methods of deepfake detection since neural networks can recognize deepfake distortions. Many papers have been published that show that CNNs work well when it comes to identifying deepfakes. For example, Zhang et al. proposed a GAN simulator that imitates artifacts created by deepfake production techniques and uses classifiers to increase their identification effectiveness. Other

Authorized licensed use limited to: Acharya Institute of Technology. Downloaded on April 13,2026 at 05:39:19 UTC from IEEE Xplore. Restrictions apply.

models include, XceptionNet as well as ResNet that performs excellent in the classification of real images from fake images [8].

A literature synthesis that included 112 papers published from 2018 through 2020 showed that deep learning techniques surpassed traditional approaches, with recognition rates exceeding 89%. Temporal features including eye blinking and head movements have also been used and models like LSTM networks have been used to look at frames sequences.

2. Classical Statistical Learning Algorithms

, but methods of classical machine learning are still important, especially when the amount of data is restricted. Based on the extracted features, the current work employed Support Vector Machines (SVM) and Random Forests for classifying deepfake content. These methods generally depends on low level features which are manually extracted and includes face shape, face texture etc to detect inconsistencies in the forged images.

In any case, the accuracy rate of classical methods is lower compared to deep learning methods with the average accuracy not exceeding 80%. This limitation makes it necessary for future works to explore better feature extraction methods, and to incorporate the strategies of deep learning [9].

3. Statistical Techniques

Statistical methods have also been employed to assess the authenticity of media. These approaches typically involve analyzing the distribution of pixel values and other statistical properties to detect anomalies indicative of manipulation. While these methods can provide valuable insights, they often lack the robustness and adaptability of machine learning techniques, particularly in the face of evolving deepfake generation methods [10].

4. Blockchain-Based Solutions

New studies have looked into the suitability of blockchain in combating deepfake and authenticating content. In other words, the use of blockchain can bring efficiency into the process of identifying the true source of digital content. This seems highly suitable for journalism and legal use, as the correctness of information is essential in these fields. Nevertheless, there are still a number of challenges that take the simple proposal of a blockchain solution and apply it to deepfake detection by comparing the chain to the work done in real life.

From the present literature, the consistent signs point to the utilization of deep learning techniques in deepfake detection because of their efficiency and malleability. Although still prominent, it is clear that problems with classical machine learning and statistical approaches explain why the detection methodologies still require further research and development. Also, the study of blockchain as a concept offers an opportunity to strengthen the idea of digital media credibility. Depending on the advancements in deepfake technology, a significant and effective method of prevention and detection of such images will also make-up a necessity [11].

A Review on State-of-the-art Deepfake Detection Methods

Due to a surge in deepfake, the need to come up with proper detection techniques has arisen. This comparative study synthesizes findings from various research papers, categorizing the detection techniques into four primary approaches given in Table 1

Table 1. Overview of different deepfake detection methods

.S.No.	Method	Overview	Performance	Strengths	Weaknesses
1	Deep Learning-Based Methods	Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have become the cornerstone of deepfake detection. These methods leverage large datasets to learn intricate patterns and features that distinguish real from manipulated media.	Studies indicate that deep learning models achieve accuracy rates exceeding 89% in detecting deepfakes. For instance, the use of architectures like XceptionNet and MesoInception-4 has shown promising results.	These methods can automatically extract features from data, making them highly effective in identifying subtle artifacts.	Their reliance on feature engineering makes them less adaptable to new types of deepfakes, and they may not perform well on complex datasets.

Authorized licensed use limited to: Acharya Institute of Technology. Downloaded on April 13,2026 at 05:39:19 UTC from IEEE Xplore. Restrictions apply.

2	Machine Learning-Based Methods	Traditional machine learning techniques, such as Support Vector Machines (SVM) and Random Forests, have been employed to detect deepfakes by analyzing specific features extracted from images and videos.	These methods generally achieve lower accuracy rates (around 70-80%) compared to deep learning approaches. They rely heavily on handcrafted features, which can limit their effectiveness.	Machine learning models are often easier to interpret and require less computational power than deep learning models.	Their reliance on feature engineering makes them less adaptable to new types of deepfakes, and they may not perform well on complex datasets.
3	Statistical Techniques	Statistical methods focus on analyzing the inherent properties of images and videos to detect anomalies indicative of manipulation.	These methods have shown mixed results, often achieving lower accuracy compared to machine learning and deep learning techniques.	Statistical techniques can be effective in specific contexts, such as identifying unique noise patterns in images.	They are generally less robust against sophisticated deepfake generation techniques and may not generalize well across different datasets.
4	Blockchain-Based Solutions	Blockchain technology offers a decentralized approach to verifying the authenticity of digital content, providing a potential solution for deepfake detection.	While still in the experimental phase, blockchain methods show promise in tracking the provenance of media.	These solutions can create immutable records of media, making it easier to verify authenticity.	The practical implementation of blockchain for deepfake detection is still underdeveloped, and challenges remain in terms of scalability and integration with existing systems.

Table 2. Comparative Summary

Detection Method	Performance	Strengths	Weaknesses
Deep Learning	>89%	High accuracy, automatic feature extraction	Requires large datasets, computationally intensive, struggles with zero-day attacks
Machine Learning	70-80%	Easier to interpret, less computationally demanding	Relies on feature engineering, less adaptable
Statistical Techniques	Mixed	Effective in specific contexts	Less robust against sophisticated deepfakes
Blockchain Solutions	Experimental	Immutable records for authenticity verification	Underdeveloped, scalability issues

Comparing with the prior and recent studies, the paper notes that, although DL-based approaches remain the main focus of deepfake research today, the development of MDFA will require the integration of machine learning, statistical analysis, and the use of such technologies as blockchain. The reason is that each of the mentioned techniques has certain advantages and disadvantages and that is why, further research on the use of different methods for deepfake detection should be aimed at combining the above approaches to further develop the capabilities of deepfake detection systems and to meet the new tasks posed by this technology.

Authorized licensed use limited to: Acharya Institute of Technology. Downloaded on April 13,2026 at 05:39:19 UTC from IEEE Xplore. Restrictions apply.

This is helpful because it provides a starting point for which future studies can commence to start to build on this from where it was left off and expand on the present work that has been presented for the detection of deepfakes.

The Role of Deep Learning for Deepfake Detection

Deep learning has become the most effective solution for the detection of deepfakes because it can learn features by itself from the massive dataset. This section briefly describes Deep Learning techniques, their topology, and their operation with respect to detecting deep fakes.

A generality regarding deep learning in deepfake detection

Specifically, Convolutional Neural Network (CNN) models are intended to work with graphical data sets. They are particularly good at working input of images and videos and are thus appropriate for detecting faked data. The main strength of deep learning is that it learns the features or hierarchies in a more complex manner therefore is able to see deeper artifacts that might denote manipulation [12].

Key Components of Deep Learning Models

1. Input Layer: The model receives raw image or video data.

2. Convolutional Layers: These layers apply filters to the input data to extract features. Each filter detects specific patterns, such as edges or textures.

3. Activation Functions: Functions like ReLU (Rectified Linear Unit) introduce non-linearity, allowing the model to learn complex relationships.

4. Pooling Layers: These layers reduce the dimensionality of the data, retaining only the most important features while discarding less significant information.

5. Fully Connected Layers: After several convolutional and pooling layers, the data is flattened and passed through fully connected layers to make predictions.

6. Output Layer: The final layer produces the classification result, indicating whether the input is a real or fake image.

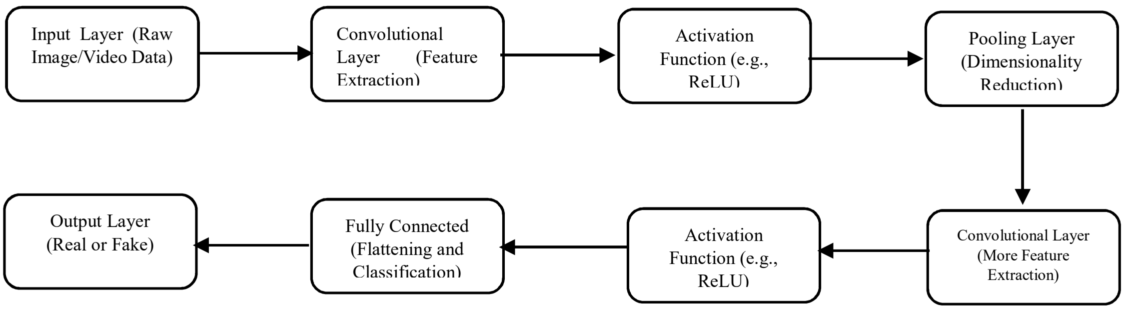

Figure 1. Deep Learning Model for Deepfake Detection

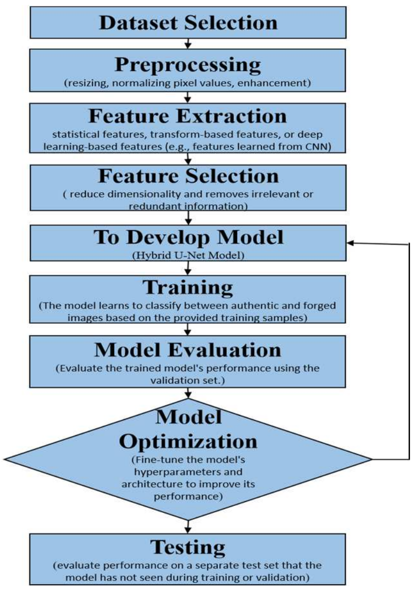

Figure 2. working flowchart of deep learning model for deepfake detection

Machine Learning-Based Methods for Deepfake Detection

Traditional ML techniques have been used to detect deepfake, especially where deep learning solutions cannot be practiced because of lack of data or lack of computational resources. They tend to use feature extraction and classification as a way of identifying manipulated content. This section describes various machine learning-based methods, the structure of the mentioned methods, and their operation in deep fake detection [13].

A Brief Introduction to Machine Learning in Deepfake Detection

Machine learning methods for deepfake detection generally involve two main steps: Two main divisions exist in our work: feature extraction and classification. Specific objectives of data analysis

Authorized licensed use limited to: Acharya Institute of Technology. Downloaded on April 13,2026 at 05:39:19 UTC from IEEE Xplore. Restrictions apply.

are to recognize some peculiar features or changes in data that could be the result of manipulation. Some of the known algorithms are Support Vector Machines (SVM), Random Forest or Decision Trees and among others [14].

Key Components of Machine Learning Models

1. Feature Extraction: This step involves identifying and extracting relevant features from the input data (images or videos). Features can include:

- Temporal Features: Changes in facial expressions or movements over time.
- Texture Features: Characteristics related to the surface quality of the image.
- Facial Landmarks: Points on the face that represent key features (e.g., eyes, nose, mouth).

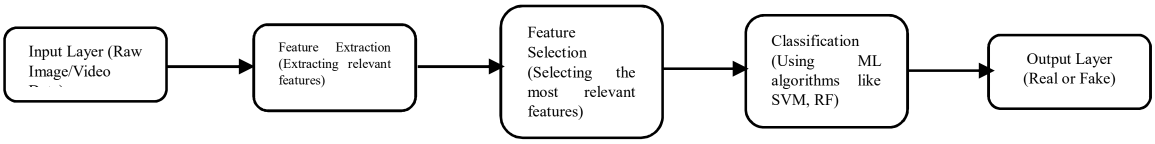

Figure 3. Machine learning model for deepfake detection

2. Feature Selection: After extraction, the most relevant features are selected to improve the model's performance and reduce dimensionality.

3. Classification: The selected features are fed into a machine learning algorithm to classify the input as either real or fake. The model is trained on labeled data, where it learns to distinguish between genuine and manipulated content.

Figure 3 illustrating the architecture of a typical machine learning model used for deepfake detection. Based on the experiments, it is figured out that it is possible to use machine learning-based methods instead of deep learning methods for deepfake detection in cases with the lack of data or computational resources. As in the case of the previous approaches, these methods generally do not provide the same level of accuracy as deep learning ones, however, they allow one to distinguish between manipulated and non-manipulated content, if specific features are used. Current studies conducted in this direction are focused on improving the stability and flexibility of machine learning algorithms to prevent the alterations in deepfake technology from displacing their effectiveness [15-16].

Blockchain-Based Deepfake Detection Method
Original Data Identification

- Media submitted to the network platform is first processed to extract a unique data fingerprint (via hashing) combined with the submitter's identity.
- A credibility value is computed using deepfake detection algorithms, assessing the likelihood of manipulation.
- The fingerprint, identity, and credibility are embedded in the media as a digital watermark using techniques like Quantization Index Modulation (QIM).
- The modified media and its metadata (e.g., hash values) are stored on the blockchain for future verification.

Deep Forgery Data Forensics and Appraisal

- **Forensic Fixation:** Media hashes and extraction processes are logged on the blockchain to ensure tamper-proof evidence.
- **Authenticity Analysis:** Media is compared with stored blockchain records to verify modifications.
- **Similarity Analysis:** Media is assessed against existing blockchain data for content similarity, using deepfake detection models where needed.
- Results are stored on-chain, ensuring transparent and immutable records.

Traceability Mechanism

- If data appears forged, its digital watermark and fingerprint are extracted for source tracing.
- The **Hamming distance** between suspicious and existing data identifiers helps determine relationships, identifying forged data and its origin.
- If no match is found, the media is treated as new and processed for initial identification.

This blockchain-based framework addresses existing challenges in deepfake detection, such as the inability to track manipulations and provide judicially recognized evidence. It integrates blockchain with forensic technologies, offering a scalable and reliable solution for combating the misuse of deepfake media [17].

Conclusion

In this study we presented various state-of-the-art methods, we presented basic techniques and

Authorized licensed use limited to: Acharya Institute of Technology. Downloaded on April 13,2026 at 05:39:19 UTC from IEEE Xplore. Restrictions apply.

discussed different detection models in this work. We summarize the overall study as follows.

- Mainly deep learning-based methods are used for deepfake detection.
- In most of the research, CNN based model is used.
- The most widely used performance metric is detection accuracy.

References

[1] Karaköse, M., Çeçen, M., & Yetiş, H. (n.d.). A New Approach for Effective Medical Deepfake Detection in Medical Images.

[2] Korshunov, P., & Marcel, S. (2018). DeepFakes: a New Threat to Face Recognition? Assessment and Detection (Version 1). arXiv. https://doi.org/10.48550/ARXIV.1812.08685

[3] Rana, S., Nobi, M. N., Sung, A. H., Shohel, M., & University of Southern Mississippi. University of Southern Mississippi. (n.d.). Deepfake Detection: A Systematic Literature Review.

[4] Reiss, T., Cavia, B., & Hoshen, Y. (2023). Detecting Deepfakes Without Seeing Any (Version 1). arXiv. https://doi.org/10.48550/ARXIV.2311.01458

[5] M. S. Rana, M. N. Nobi, B. Murali and A. H. Sung, "Deepfake Detection: A Systematic Literature Review," in *IEEE Access*, vol. 10, pp. 25494-25513, 2022, doi: 10.1109/ACCESS.2022.3154404

[6] S. Dasgupta, J. Mason, X. Yuan, O. Odeyomi, and K. Roy, "Enhancing Deepfake Detection using SE Block Attention with CNN," *2024 International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD)*, pp. 1-6, 2024.

[7] B. Ghita, I. Kuzminykh, A. Usama, T. Bakhshi, and J. Marchang, "Deepfake Image Detection Using Vision Transformer Models," *2024 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*, pp. 332-335, 2024.

[8] V. Rogovoi, V. M. Korzhuk, and O. A. Kokorina, "Development of a Deepfake Detection Method: Application of Frequency Analysis and Reduction of the Image Color Space to Improve Classification Accuracy," *2024 V International Conference on Neural Networks and Neurotechnologies (NeuroNT)*, pp. 36-39, 2024.

[9] Y. Yang, N. B. Idris, D. Yu, C. Liu, and H. Wu, "Decentralized Deepfake Task Management Algorithm Based on Blockchain and Edge Computing," *IEEE Access*, vol. 12, pp. 86456-86469, 2024.

[10] M. S. Rana, M. N. Nobi, and A. Sung, "DeepDistAL: Deepfake Dataset Distillation using Active Learning," *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 7723-7730, 2024.

[11] Y. S. Taspinar and I. Cinar, "Distinguishing Between AI Images and Real Images with Hybrid Image Classification Methods," *2024 13th Mediterranean Conference on Embedded Computing (MECO)*, pp. 1-4, 2024.

[12] S. Antad, V. V. Arthamwar, R. K. Deshmukh, A. U. Chame, and H. P. Chhangani, "A Hybrid Approach for Deepfake Detection using CNN-RNN," *2024 OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 4.0*, pp. 1-6, 2024.

[13] M. A. P. Putra, N. W. Utami, I. G. J. E. Putra, N. Karna, A. Zainudin, and G. A. R. Sampedro, "Collaborative Decentralized Learning for Detecting Deepfake Videos in Entertainment," *2024 IEEE Gaming, Entertainment, and Media Conference (GEM)*, pp. 1-4, 2024.

[14] S. K. Ahir and O. M. Adedayo, "Multimedia Forensics: Preserving Video Integrity with Blockchain," *2024 12th International Symposium on Digital Forensics and Security (ISDFS)*, pp. 1-6, 2024.

[15] R. Singh, K. Ashwini, B. C. Priya, and K. P. Kumar, "Deepfake Face Extraction and Detection Using MTCNN-Vision Transformers," *2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE)*, pp. 1-8, 2024.

[16] A. Gadde, G. D. K. Kishore, T. Talari, S. L. Nunna, R. C. Nannapaneni, and K. M. S. K. Vamsi, "Detecting Deepfake Images: A Deep Learning Approach with Streamlit Integration," *2024 International Conference on Science Technology Engineering and Management (ICSTEM)*, pp. 1-7, 2024.

[17] M. Priya, J. Murugesan, P. Bhuvaneswari, M. Rubigha, S. Lalithambikai and B. Mohanraj, "Preserving Visual Authenticity: Block chain-Augmented AI Frameworks for Advanced Digital Deception Recognition and Mitigation," *2024 5th International Conference on Smart Electronics and Communication (ICOSEC)*, Trichy, India, 2024, pp. 707-713, doi:10.1109/ICOSEC61587.2024.10722740.

Authorized licensed use limited to: Acharya Institute of Technology. Downloaded on April 13,2026 at 05:39:19 UTC from IEEE Xplore. Restrictions apply.